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ORIGINAL PAPER

Dual-salts of LiTFSI and LiODFB for high voltage cathode $LiNi_{0.5}Mn_{1.5}O_4$

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Abstract The $LiNi_{0.5}Mn_{1.5}O_4$ has been investigated as promising because of its high operating voltage (4.7 V vs. Li^+/Li) and decent specific capacity. Despite these advantages, however, the market penetration of this material was significantly hindered by the lack of proper electrolytes that can work stably up to 5 V. To achieve better cycling stability, mixed salts of lithium bis (trifluoromethanesulfonylimide) (LiTFSI) and lithium difluoro (oxalate) borate (LiODFB) are used in $LiNi_{0.5}Mn_{1.5}O_4$-based batteries to replace the lithium hexafluoride phosphate ($LiPF_6$) salt in non-aqueous electrolytes. The SEM and CV data indicate that the corrosion to Al current collector caused by LiTFSI-based electrolytes under high voltage (~5 V vs. Li^+/Li) can be largely reduced by adding LiODFB into LiTFSI-based electrolytes. And a boron-based passivation layer from the oxidative decomposition of LiODFB to suppress the corrosion based on the XPS tests results. In addition, the $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte has better cycling stability and rate capability for $LiNi_{0.5}Mn_{1.5}O_4$-based cells than $LiPF_6$-based electrolyte.

Keywords Lithium-ion batteries · Dual-salts electrolyte · Aluminum corrosion · Cycling stability

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Introduction

In recent years, lithium-ion batteries with high energy density has attracted much attention due to the urgent demand of energy storage technologies for electric vehicles (EV), hybrid electric vehicles (HEV), and smart grids [1]. In order to increase the energy density, a general approach is to use cathode materials with both high operating voltages (~5 V vs. Li^+/Li) and specific capacity [2]. Accordingly, the $LiNi_{0.5}Mn_{1.5}O_4$ has been investigated as promising because of its high operating voltage (4.7 V vs. Li^+/Li) and decent specific capacity [3]. Despite these advantages, however, the market penetration of this material was significantly hindered by the lack of proper electrolytes that can work stably up to 5 V [4]. Currently, commercial high voltage electrolytes are mainly based on the lithium hexafluoride phosphate ($LiPF_6$), which however is not very stable and may decompose into lithium fluoride (LiF) and phosphorus pentafluoride (PF_5) [5, 6]. The generated PF_5 reacts with even trace amount of water to produce hydrogen fluoride (HF) [7]. The formed HF is very reactive and corrodes not only the solid electrolyte interphase (*SEI*) layers on graphite anodes but also cathode materials such as Li transition metal oxides. The end result is dissolution of cathode, interfacial failure of anode,

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and thus fast capacity degradation of the battery [8]. The above cell failure induced by $LiPF_6$ decomposition may be exacerbated under conditions of high voltage (~5 V vs. Li^+/Li) or at elevated operation temperatures. Therefore, the alternative electrolytes with much improved stability are urgently needed for practical applications of the $LiNi_{0.5}Mn_{1.5}O_4$ cathode materials.

We previously [10] found that the LiTFSI-LiODFB-based (LiTFSI, lithium bis (trifluoromethanesulfonylimide), $LiN(CF_3SO_2)_2$; LiODFB, lithium difluoro (oxalate) borate, $(LiBC_2O_4F_2)$) electrolyte can effectively improve the electrochemical performance of LiFePO4 cathode compared to that with $LiPF_6$-based electrolyte, indicating that LiTFSI is a promising Li salt for electrolyte with high stability against water and heat [9]. However, obvious Al corrosion with LiTFSI-based electrolyte is usually identified at voltages higher than 3.7 V vs. Li/Li^+, which is undesired for practical application of Li-ion batteries [8]. Interestingly, it was found that adding a small amount of LiODFB into the LiTFSI electrolyte could effectively suppress the observed Al corrosion [10]. So it is a reasonable speculation that the electrolyte containing both LiTFSI and LiODFB may improve electrochemical window of the electrolyte and at the same time suppress Al corrosion.

In this work, a novel electrolyte with dual-salts of LiTFSI and LiODFB was successfully prepared and evaluated using $LiNi_{0.5}Mn_{1.5}O_4$ as the cathode. And the electrochemical performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ batteries using the dual-salts of LiTFSI and LiODFB was investigated. We found that under high voltage (~5 V vs. Li^+/Li), adding an appropriate amount of LiODFB into the LiTFSI electrolyte could effectively reduce aluminum corrosion and improve electrochemical performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells.

Experimental

Cell preparation

EC (ethylene carbonate), EMC (ethyl methyl carbonate), and LiTFSI of battery grade were purchased from Guangzhou Tinci Materials Technology Co. Ltd. China. LiODFB of battery grade was ordered from Beijing Isomersyn Technology Co. Ltd. China. The aluminum foil and coin cell cases (EQ-CR2025-CASE button cells case, made of 304 stainless steel with sealing O-ring) were purchased from Shenzhen Keijing Star Technology Co. Ltd. China. The $LiNi_{0.5}Mn_{1.5}O_4$ cathode material was obtained from Hunan Shanshan Advanced Materials Co. Ltd. China.

The cathode slurry was prepared by the mixing of 80 wt.% $LiNi_{0.5}Mn_{1.5}O_4$, 10 wt.% Super-P, and 10 wt.% PVDF, which was then coated on Al foil. Half cells with $LiNi_{0.5}Mn_{1.5}O_4$ cathode, Celgard 2400 polypropylene film separator, electrolyte, and Li foil anode were assembled in an argon-filled glove box (LAB2000, Etelux) with moisture and O_2 under 1 ppm. The solvent EC was first mixed with EMC with a weight ratio of 4:6, then 1 M $LiTFSI_x$-$LiODFB_{(1-x)}$ (x is the molar ratio of LiTFSI in the LiTFSI-LiODFB mixed salts) were dissolved in the above solution. The control electrolyte of 1 M $LiPF_6$ in EC-EMC solution was also prepared and used for comparison.

Measurement

The cyclic voltammogram (CV) was performed on LK2005A electrochemical workstation (Tianjin, China) with a scan rate of 5.0 mV s^{-1} and a voltage range from 2.0 to 6.0 V. Ionic conductivity of electrolyte was measured by a conductivity meter (Shanghai REX Instrument Factory, DDS-307 A). The morphology of Al foil surface before and after charging was observed by field emission scanning electron microscope (FESEM, Hitachi Limited, SU8010). The elemental analysis of the Al foil surface film was carried out by X-ray photoelectron spectroscopy (XPS, Kratos AXIS ULTR^{DLD}). Electrochemical impedance spectroscopy testing was conducted on a PARSTAT MC 2000 (USA). The cell performances of Li/$LiNi_{0.5}Mn_{1.5}O_4$ were tested on a LAND CT2001A battery test system (Wuhan, China) at various current densities between 3.0 and 4.95 V (vs. Li/Li^+). All of the tests were conducted at room temperature (20 °C).

Results and discussion

Cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in $LiPF_6$ and LiTFSI-based electrolytes

Figure 1 shows the cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ batteries in the electrolyte based on single-salt ($LiPF_6$ or

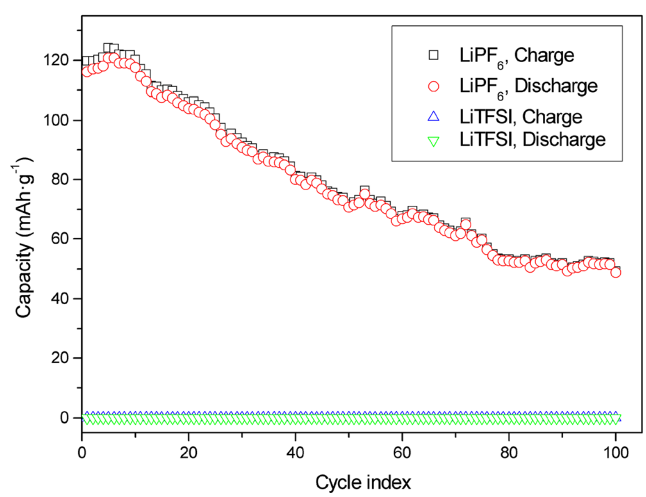

Fig. 1 Cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in the single-salt electrolytes ($LiPF_6$ and LiTFSI) at room temperature. The charge-discharge current rate was 0.5 C between 3.0 and 4.95 V, where the first two formation cycle data at the 0.1 C rate are not shown

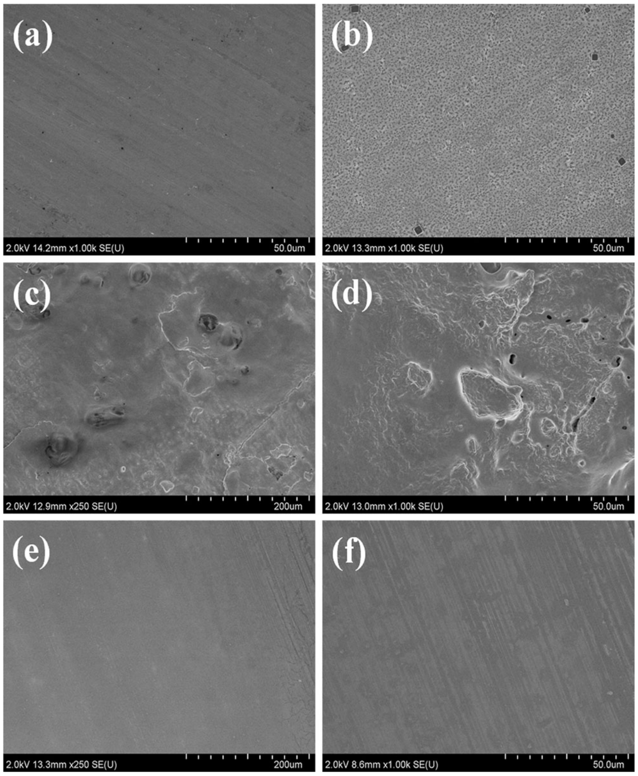

Fig. 2 SEM micrographs of (**a**) pure Al foil and Al foils after corrosion test at 4.95 V for 1 week in 1 M $LiPF_6$-based electrolyte (**b**), 1 M LiTFSI-based electrolyte (**c**, **d**), and $LiTFSI_{0.6}$-$LiODFB_{0.4}$ electrolyte (**e**, **f**)

LiTFSI). The capacity of the cell with $LiPF_6$-based electrolyte is not stable and decays very fast upon cycling. The observed fast capacity degradation may be ascribed to oxidation/decomposition of electrolyte solvent under the high voltage as 5 V [11]. In addition, the $LiPF_6$ itself can also decompose into LiF and PF_5 at high voltages. The formation of the LiF will reduce both conductivity and numbers of free Li^+ ions, increasing the cell impedance and leading to capacity fading. On the other hand, the HF produced by reaction of PF_5 with H_2O will corrode $LiNi_{0.5}Mn_{1.5}O_4$ cathode materials to accelerate the capacity fading [12]. The trend of $LiPF_6$ decomposition may be further enhanced under high voltages (~5 V vs. Li^+/Li). In the LiTFSI-based electrolyte, there is no HF generated because LiTFSI has high stability and good tolerance to water [13]. However, no capacity was obtained for the cell with LiTFSI-based electrolyte. This may be caused by Al corrosion, which produces thick layer of Al $(TFSI)_3$ to isolate cathode material with Al current collector [14–16].

Aluminum foil corrosion experiment

It was found that adding a small amount of LiODFB into the LiTFSI electrolyte could effectively suppress aluminum corrosion [10]. In order to verify whether the LiODFB could reduce the corrosion of Al in LiTFSI-based electrolytes under high voltage (~5 V vs. Li^+/Li), corrosion behaviors of Al within various electrolytes were investigated. Three electrolytes of 1 M $LiPF_6$, 1 M LiTFSI, and 1 M $LiTFSI_{0.6}$-$LiODFB_{0.4}$ in EC-EMC solutions were studied and compared. Figure 2 shows the SEM images of Al foil before and after the corrosion test, and significant changes can be identified. When the pure LiTFSI electrolyte was used, the fresh and smooth Al foil (Fig. 2a) was seriously corroded and even broke into pieces (Fig. 2c, d). While in the 1 M $LiTFSI_{0.6}$-$LiODFB_{0.4}$ electrolyte, the Al foil remained similar morphology and no obvious signs of corrosion were observed (Fig. 2e, f). Compared to 1 M LiTFSI electrolyte, passivation film could be observed on the surface of Al foil in 1 M $LiPF_6$ electrolyte (Fig. 2b). The main ingredients of passivation film are AlF_3, $Al_2(CO_3)_3$, and $Al(PF_6)_3$, which originates from reactions of $LiPF_6$ and Al [17].

The XPS analysis of Al foil in 1 M $LiTFSI_{0.6}$-$LiODFB_{0.4}$ electrolyte demonstrates the existence of element B as well as other elements such as O, F, S, and N (Fig. 3a). Those elements are from the decomposition of both LiODFB and LiTFSI. The core XPS spectrum of B indicates the presence of B_2O_3 corresponding to the binding energy of 193.4 eV

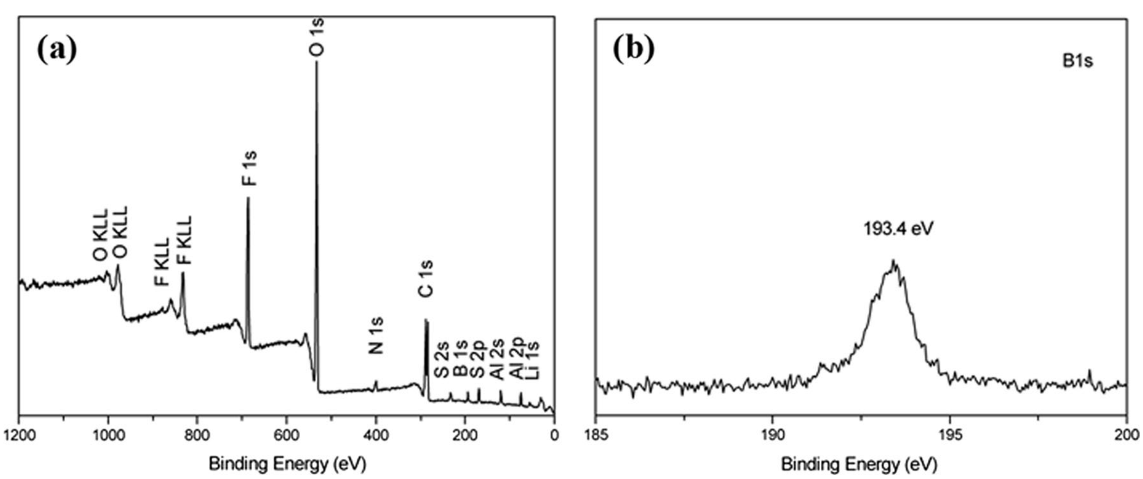

Fig. 3 The wide-survey XPS spectra of the Al foil from the cell using 1 M $LiTFSI_{0.6}$-$LiODFB_{0.4}$ electrolyte (**a**) and the core XPS spectra of B (**b**)

(Fig. 3c). It was reported that the B-O compounds from LiBOB can protect the Al foil from corrosion by LiTFSI-based electrolyte when LiBOB was added to LiTFSI-based electrolyte [8]. LiODFB has similar structure with LiBOB and thus functions in a similar manner to prevent Al foil from corrosion by forming a boron-based passivation layer.

The CV results (Fig. 4) reveal a significant decrease in the corrosion current density when LiODFB was added to the LiTFSI-based electrolytes. The corrosion current density decreases from about 1.7 mA cm^{-2} (Fig. 4a) for the 1 M LiTFSI-based electrolyte to about 0.65 mA cm^{-2} when LiODFB was added at 0.2 M (Fig. 4c). When the content of LiODFB was

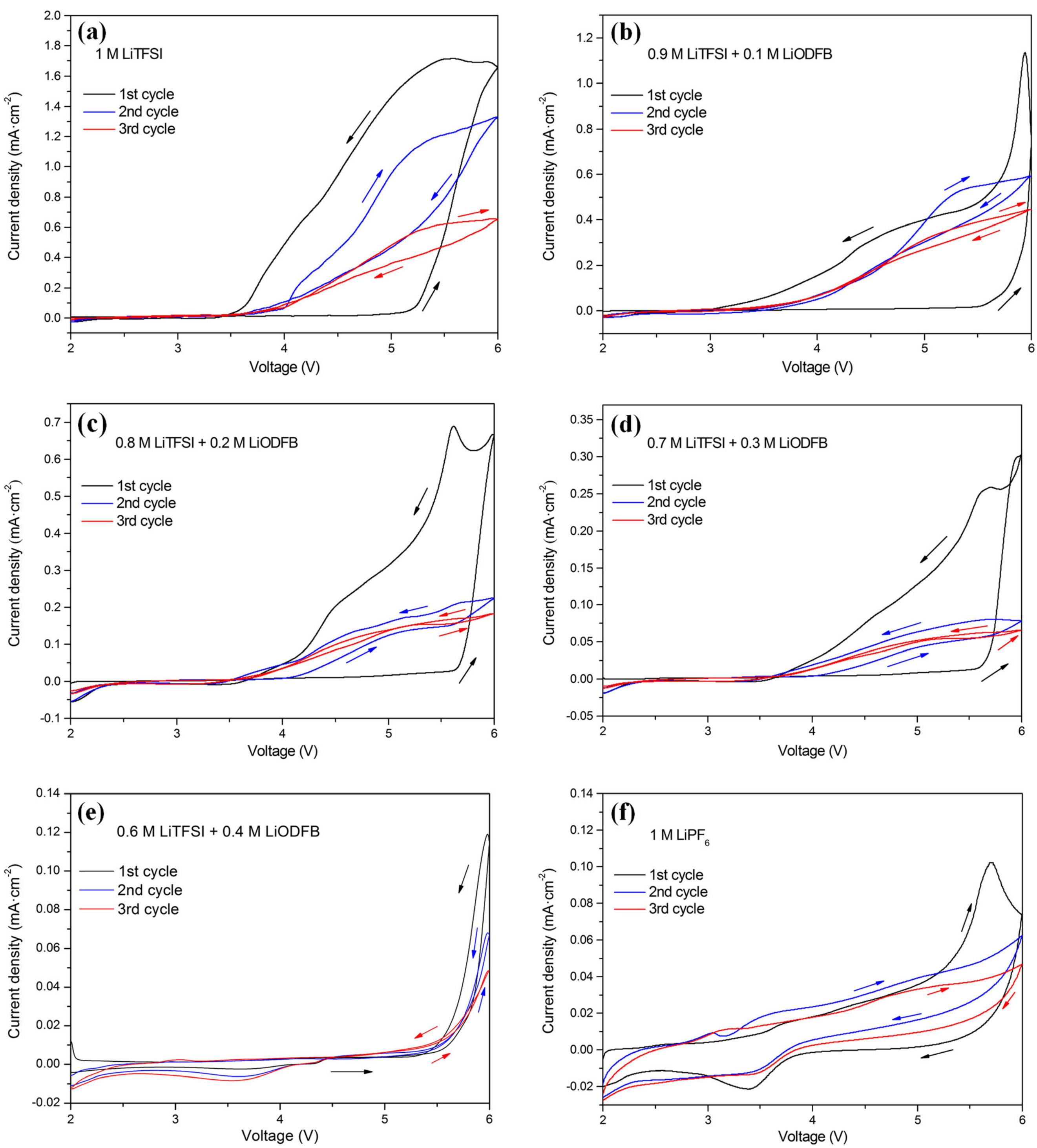

Fig. 4 CV curves of Al electrodes in different electrolytes, 1 M LiTFSI (**a**), 1 M $LiTFSI_{0.9}$-$LiODFB_{0.1}$ (**b**), 1 M $LiTFSI_{0.8}$-$LiODFB_{0.2}$ (**c**), 1 M $LiTFSI_{0.7}$-$LiODFB_{0.3}$ (**d**), 1 M $LiTFSI_{0.6}$-$LiODFB_{0.4}$ (**e**), and 1 M $LiPF_6$ (**f**)

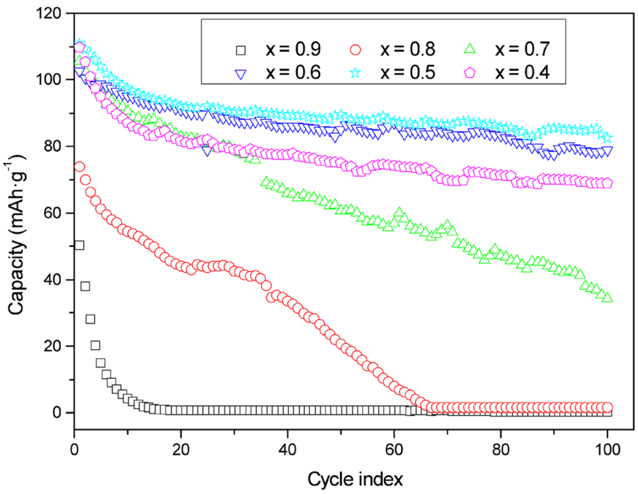

Fig. 5 Cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells using 1 M $LiTFSI_x$-$LiODFB_{(1-x)}$ at room temperature. The charge-discharge current rate was 0.5 C between 3.0 and 4.95 V, where the first two formation cycle data at the 0.1 C rate are not shown

increased to 0.4 M (Fig. 4e), corrosion is suppressed completely. Actually, the corrosion current density has decreased to the same level as that in the $LiPF_6$-based electrolyte (Fig. 4f). These results confirm that LiODFB can effectively suppress the corrosion of Al foil in LiTFSI-based electrolytes under the high voltage as 5 V, which is in accordance with previous report that LiODFB has excellent ability to passivate Al at high potentials [18, 19].

Cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in 1 M $LiTFSI_x$-$LiODFB_{(1-x)}$-based electrolytes

Figure 5 shows the long-term cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells using 1 M $LiTFSI_x$-$LiODFB_{(1-x)}$-based electrolyte. The cells were charged-discharged at a current rate of 0.5 C between 3.0 and 4.95 V. The improved cycling stability was observed with adding of LiODFB, which agrees well with the CV test results as discussed above. With increase of LiODFB concentrations from 0.3, 0.4, to 0.5 M, corresponding cycling stability was also increased. This indicates that a high concentration of LiODFB is essential to form a compact passivation layer to protect Al current collector. However, as shown in Table 1, the conductivity of the LiTFSI-LiODFB electrolytes keeps decreasing with increasing LiODFB concentration, from 7.712 ms cm^{-1} for $x = 0.9$, to 7.516 ms cm^{-1} for $x = 0.8$, 7.468 ms cm^{-1} for $x = 0.7$, 7.372 ms $cm^{-1}$for $x = 0.6$, 7.092 ms cm^{-1} for $x = 0.5$, and 6.967 ms cm^{-1} for $x = 0.4$. This means the presence of LiODFB in the electrolyte leads to low ionic conductivity of electrolyte, which may lead to capacity drop if concentration of LiODFB is too high as reflected in the cell performance in Fig. 5. The best cycling stability performance is obtained with LiTFSI/LiODFB = 1/1.

The performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in $LiPF_6$ and $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte

Figure 6 compares long-term cycling performance of Li/$LiNi_{0.5}Mn_{1.5}O_4$cells with 1 M $LiTFSI_{0.5}$-$LiODFB_{0.5}$ and 1 M $LiPF_6$. It was found that Li/$LiNi_{0.5}Mn_{1.5}O_4$ cell with 1 M $LiTFSI_{0.5}$-$LiODFB_{0.5}$ electrolyte shows high first discharge capacity of 110.7 mAh g^{-1} and retains a capacity of 82.4 mAh g^{-1} after the 100 cycles, with a capacity retention ratio of 74.4 %. While with 1 M $LiPF_6$ electrolyte, a quick capacity fading was observed from 116.0 mAh g^{-1} at first discharge to 48.7 mAh g^{-1} at 100th cycle, with a capacity retention of 42.0 %. This further demonstrates that adding LiODFB into LiTFSI-based electrolytes can effectively suppress Al corrosion [10].

The improved electrolyte stability as well as suppressed Al corrosion makes it possible to further evaluate rate capability of $LiNi_{0.5}Mn_{1.5}O_4$ cells, which is critical for high power applications in electric vehicles.

As shown in Fig. 6b, negligible difference was observed between the discharge capacities of the cells using the $LiPF_6$ and $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolytes at low current rates (0.1, 0.2, and 0.5 C). When current rate increases to 1 from 0.5 C, the discharge capacity of the cell using the $LiPF_6$-based electrolyte drops from 113.6 to 97.3 mAh g^{-1}, but the cell with the $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte remains 103 mAh g^{-1}. At 2 C rate, the cell with the $LiTFSI_{0.5}$-$LiODFB_{0.5}$ electrolyte has a capacity retention of 69.7 % (125.2 and 87.3 mAh g^{-1} at 0.1 and 2 C, respectively); while under the same condition, the $LiPF_6$-based cell shows a capacity retention of 64.3 % (125.1 and 80.5 mAh g^{-1} for 0.1 and 2 C, respectively). After the current was decreased to low current rate of 0.2 C after rate testing, the discharge capacities of both the cells can be recovered. The rate advantage of the $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte is attributed to the excellent stability of both the LiTFSI and LiODFB salts and the lower acid concentration than in the $LiPF_6$-based [9, 13].

Figure 6c shows the voltage profiles of the Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in two kinds of electrolytes at the 50th

Table 1 Conductivity of 1 M l$LiTFSI_x$-$LiODFB_{(1-x)}$ in EC-EMC (4:6 wt.%)

Concentration of LiODFB (mol L^{-1})	0	0.1	0.2	0.3	0.4	0.5	0.6
Conductivity (ms cm^{-1})	7.918	7.712	7.516	7.468	7.372	7.092	6.967

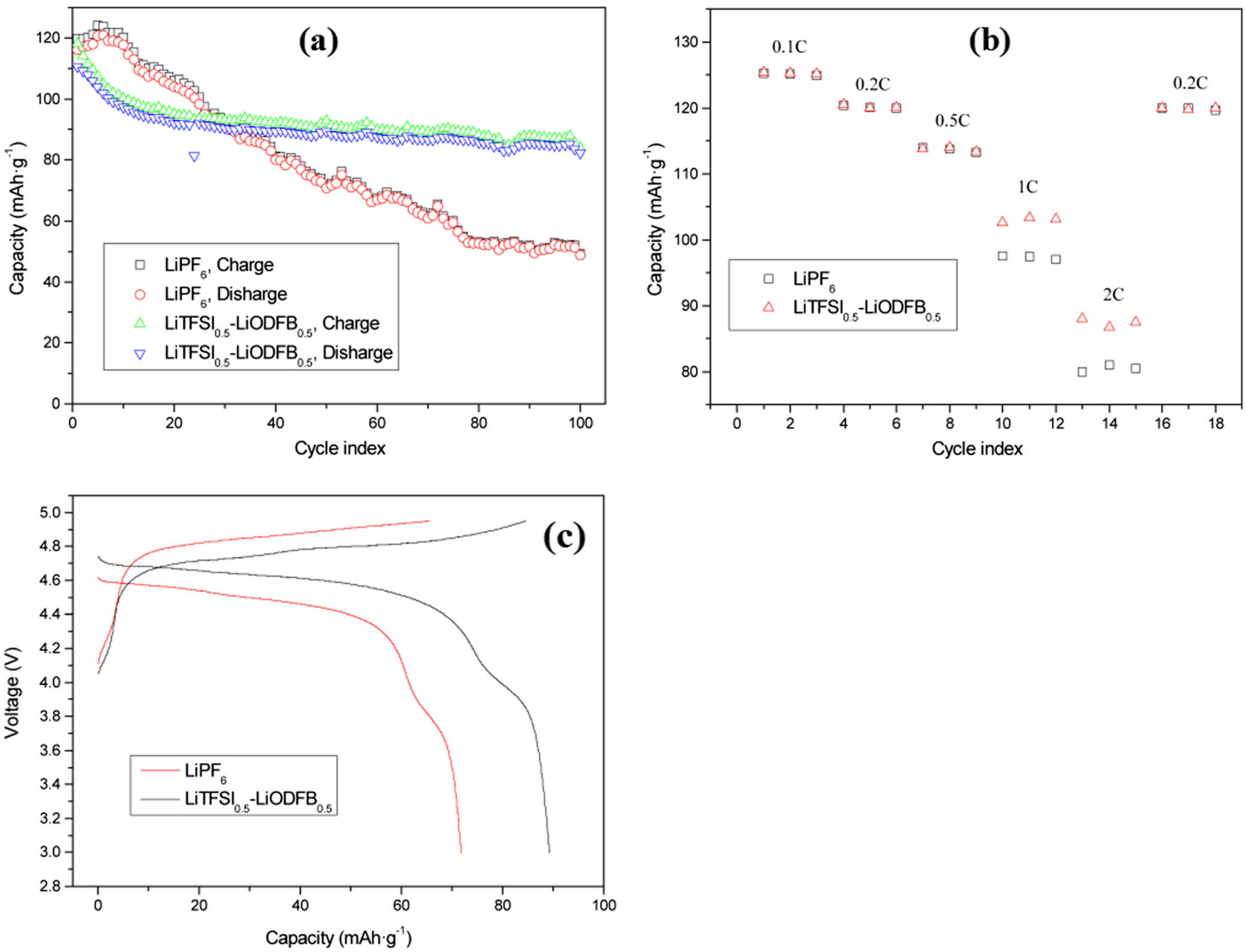

Fig. 6 The cycling performance (**a**), discharged rate capability (**b**), and voltage profile (**c**) of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in 1 M $LiPF_6$ and 1 M $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte at room temperature

cycle recorded at 0.5 C. Besides lower electrochemical polarization during the charge-discharge processes, the cell with $LiTFSI_{0.5}$-$LiODFB_{0.5}$ electrolyte shows higher discharge capacity than that with $LiPF_6$ electrolyte. This can be attributed to the excellent stability of $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte in Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells.

The impedance analysis of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells with two kinds of electrolytes

Figure 7 shows the impedance spectra of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells in different electrolytes after different cycles (a, b) original data and (c, d) fitted data. The high frequency range represents the solid electrolyte interphase film resistance (R_{SEI}), the middle frequency range can be ascribed to the charge transfer resistance (R_{ct}), and the low frequency region corresponds to the lithium ions diffusion (W_{dif}) [20, 21]. It can be seen that the impedance spectra of the cell using $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte shows two obvious semicircle, but the cell using $LiPF_6$-based electrolyte shows only one semicircle. Because, in many $LiPF_6$-based cases, the separation among the above major processes and time constants is not clear, so the high-to-medium frequency spectra may appear as a single flat semicircle, which reflects both Li-ion migration through surface films and interfacial charge transfer. Kinetic parameters of Li/$LiNi_{0.5}Mn_{1.5}O_4$ cells using $LiPF_6$ or $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolytes at different cycles are shown in Table 2. Obviously, the cell using $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte shows lower R_{SEI} + R_{ct} than the cell using $LiPF_6$-based electrolyte. One of the reasons is the formation of an effective SEI film by LiODFB-based electrolyte. Another primary reason could be the good compatibility between $LiNi_{0.5}Mn_{1.5}O_4$ active materials and the mixed salts electrolyte. Although the impedances increase with cycle numbers for two different cells, the cell using $LiTFSI_{0.5}$-$LiODFB_{0.5}$-based electrolyte shows a relatively less dramatic enlargement, which could be consistent

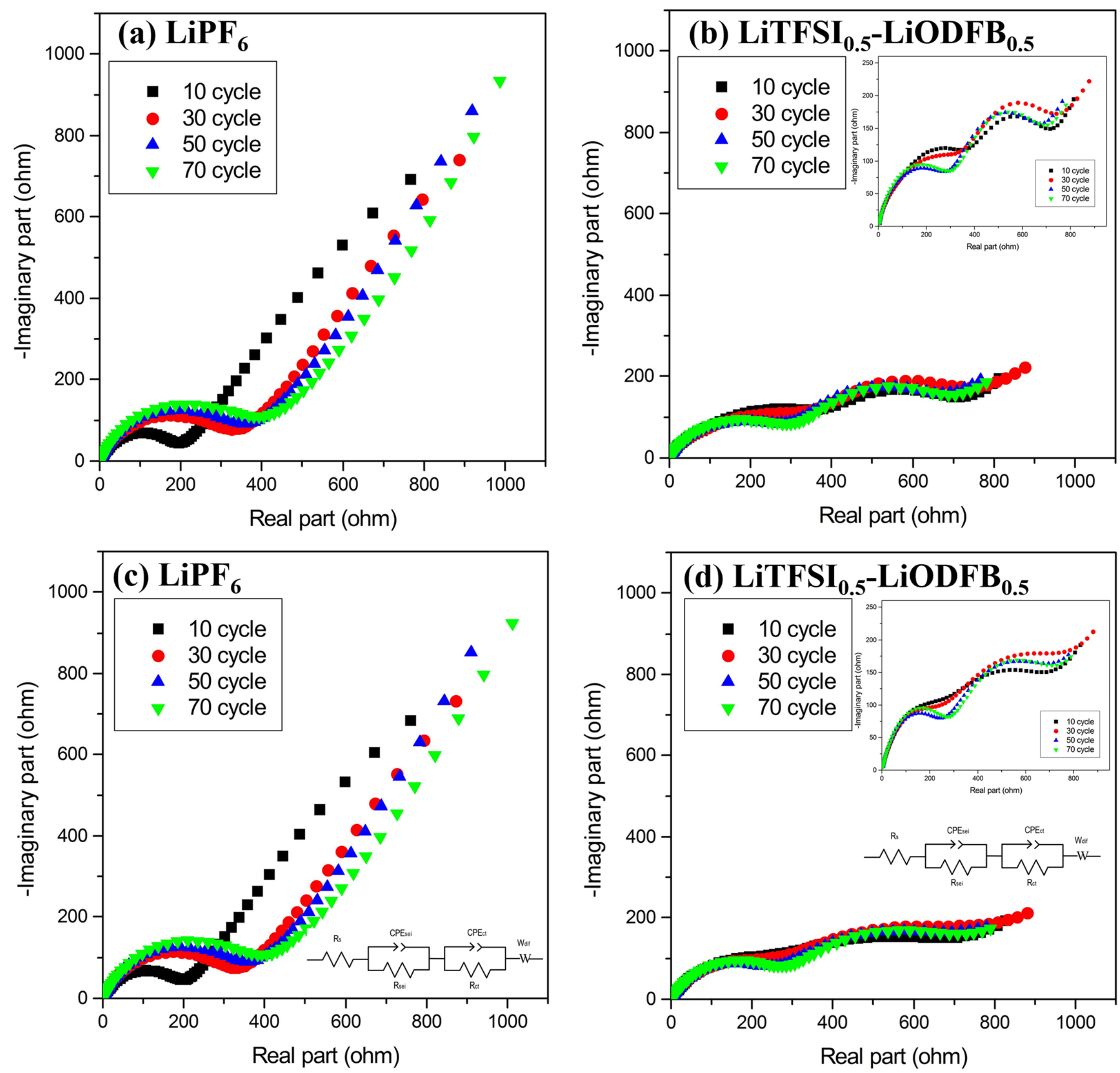

Fig. 7 The impedance spectra of Li/LiNi$_{0.5}$Mn$_{1.5}$O$_4$ cells in different electrolytes after different cycles original data (**a**, **b**) and fitted data (**c**, **d**)

with its good capacity retention, high rate capability, and lower polarization.

Table 2 Kinetic parameters of Li/LiNi$_{0.5}$Mn$_{1.5}$O$_4$ batteries using LiPF$_6$ or LiTFSI$_{0.5}$-LiODFB$_{0.5}$-based electrolytes at different cycles

Electrolyte	Cycle	R_s (Ω)	R_{sei}/R_{ct} (Ω)		W_{dif} (Yo)
LiPF$_6$	10	1.484	837.6	184.4	0.0023
	30	1.02	819.7	304.9	0.0018
	50	1.077	48,885	330.0	0.0016
	70	1.568	1.8*10^{11}	360.0	0.0014
LiTFSI$_{0.5}$-LiODFB$_{0.5}$	10	0.4529	238.6	475.8	0.0059
	30	0.8135	231.7	555.7	0.0058
	50	1.384	454	250.4	0.0074
	70	1.688	414.7	280.9	0.0072

Conclusions

The synergistic effects of the mixed salts of LiTFSI and LiODFB for high voltage (~5 V vs. Li$^+$/Li) Li/LiNi$_{0.5}$Mn$_{1.5}$O$_4$ cells were investigated this paper. Compared with Li/LiNi$_{0.5}$Mn$_{1.5}$O$_4$ cells using LiPF$_6$-based electrolyte, the cells with LiTFSI$_{0.5}$-LiODFB$_{0.5}$-based as electrolyte have excellent cycling stability and rate performance. Adding LiODFB as a co-salt can effectively reduce the Al foil corrosion from LiTFSI-based electrolytes at high voltage condition (~5 V vs. Li$^+$/Li). In brief, this study shows mix lithium salts could have great potential for high voltage Li-ion batteries.

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